ARITHMETIC SELECTION AND CATALAN GEOMETRY OF THE MERTENS WALK: EXACT STRUCTURES, CAPACITY DEFICITS, OCCUPATION ENERGY, AND OPEN ROUTES

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Abstract. We organize several exact and experimental structures around the Möbius function $\mu(n)$ and its summatory function $M(x)$. The squarefree-compressed Mertens walk is an exact nearest-neighbour path; complete zero-to-zero excursions are primitive Dyck paths and open pieces are ballot meanders. This produces a canonical Catalan state space, but ambient Catalan rarity alone does not restrict the arithmetic selector. We therefore emphasize arithmetic mechanisms that can act on the whole prefix profile.

The first part develops the occupation-energy formulation $\mathcal{E}_q(x,H)=\sum_{h\le H}|M(x+h)-M(x)|^{2q}$ and the exact quadratic operator $J_H^*J_H$. The all-q dyadic hierarchy is an exact RH criterion, while any fixed power saving over the ballistic quadratic scale is a strictly weaker and quantitatively meaningful first target. The second part combines adversarial multiplicative completions, signed displacement cuts, and collective Catalan capacities. For a finite squarefree prime product P, the squarefree integers decompose into components $aD(P)$, and the exact worst-case prefix capacity is expressed by an L^1 norm of a signed divisor-window kernel. Adding a prime acts by a delayed anti-copy, gives a monotone capacity ratio, and also yields a lower barrier showing a tradeoff between confinement and remaining entropy. A further no-go theorem shows that static Ramaré relations over a finite prime set do not couple different $aD(P)$ components; genuine inter-component gain must use causal truncation or an analytic recombination estimate.

The synthesis also retains the signed-cut extinction programme, the ordinal-prime transform and its coercive negative-excursion energy, the divisibility-surface and p-adic shell coordinates, and a one-way Catalan calibration of hypothetical off-critical zeta zeros. Finite computations through $N = 455{,}052{,}511$ are recorded only as experiments. The resulting research frontier is explicit: obtain a proved, non-tautological, sign-sensitive collective restriction that reduces high Catalan sectors or occupation energy by a fixed power while leaving substantial arithmetic freedom.

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Remark 0.1 (Status and purpose). This document is a research synthesis and working notebook. It consolidates the strongest exact constructions, negative results, finite experiments, and open mechanisms developed in a collection of Möbius–Mertens manuscripts and project notebooks. The labels [LIT], [DED], [COMP], [OPEN], [NEG], and [HEUR]mean, respectively: established literature input; rigorous deduction recorded in the project; finite computation; unresolved target; rigorous or structural negative result; and heuristic. No literature-wide priority claim is made for deductions marked [DED]without a separate bibliographic audit. In particular, exact reformulations equivalent to the Riemann hypothesis are not presented as progress toward its proof unless an independent arithmetic estimate is supplied.

1. Main objects and logical boundary

Let

$$M(x) = \sum_{n \le x} \mu(n), \tag{1.1}$$

and let

$$Q(x) = \sum_{n \le x} \mu(n)^2 = \#\{n \le x : n \text{ is squarefree}\}. \tag{1.2}$$

The standard squarefree counting asymptotic is

$$Q(x) = \frac{6}{\pi^2} x + O(\sqrt{x}). \tag{1.3}$$

The classical Mertens criterion states that

$$\text{RH} \iff M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \quad \text{for every } \varepsilon > 0; \tag{1.4}$$

see, for example, [25]. The obsolete Mertens conjecture $|M(x)| < \sqrt{x}$ is false [21]; it must never be used as an asymptotic surrogate for (1.4).

Warning 1.1 (Criterion versus mechanism). Any construction that immediately reduces to (1.4), to a zero-free half-plane for $1/\zeta$, or to an equivalent mean-square statement is a criterion or change of coordinates. It becomes a proof route only after a genuinely independent arithmetic estimate is established.

The central methodological boundary used throughout this notebook is

(1.5)

$$\boxed{\text{partial arithmetic information} \Longrightarrow \text{uniform non-tautological geometric or energetic inequality.}}$$

The left side must not contain the desired Mertens bound in disguised form.

2. The squarefree-compressed Mertens walk

Let

$$1 = q_1 < q_2 < q_3 < \cdots \tag{2.1}$$

be the squarefree integers in increasing order, and define

$$X_m = \sum_{j \le m} \mu(q_j). \tag{2.2}$$

Proposition 2.1 (Exact compression [DED])**.** *For every $m \ge 1$ and real $x \ge 1$,*

$$X_m = M(q_m), \tag{2.3}$$

$$M(x) = X_{Q(x)}. \tag{2.4}$$

Proof. The terms deleted from ordinary time are exactly the terms for which $\mu(n) = 0$. Thus the nonzero increments are unchanged and only holding times are removed. □

Let

$$0 = z_0 < z_1 < z_2 < \cdots \tag{2.5}$$

be successive return times of X_m to zero, when they exist. Put

$$L_r = z_r - z_{r-1} = 2n_r. \tag{2.6}$$

Parity forces each complete excursion length to be even.

Proposition 2.2 (Primitive Catalan factorization [DED])**.** *Fix a finite return skeleton and the sign of each complete excursion. The number of nearest-neighbour paths realizing excursions of lengths $2n_1, \dots, 2n_R$ is*

$$\mathcal{N}(z_1, \dots, z_R) = \prod_{r=1}^{R} C_{n_r - 1}, \tag{2.7}$$

where

$$C_n = \frac{1}{n+1}\binom{2n}{n} \tag{2.8}$$

is the nth Catalan number.

Proof. After reflection of a negative excursion, removing the first up-step and final down-step gives an arbitrary Dyck path of semilength $n_r - 1$. Distinct complete excursions are combinatorially independent once the skeleton and signs are fixed. □

For open pieces the exact ballot count is equally important.

Proposition 2.3 (Strict meander endpoint count [LIT])**.** *Let $t \ge 1$ and $h > 0$ with $t \equiv h \pmod 2$. The number of ± 1 paths from 0 to h in t steps that remain strictly positive after time zero is*

$$\#\mathcal{M}_{t,h} = \frac{h}{t}\binom{t}{(t+h)/2}. \tag{2.9}$$

The ordinary-time embedding is retained by the holding times of the squarefree clock. For a window $(a, b]$, let

$$w_j(a, b) = \#\{n \in (a, b] : Q(n) = j\}. \tag{2.10}$$

Then any ordinary-time occupation statistic can be written as a weighted statistic of the compressed path. Deleting zeros therefore preserves the path shape but not the physical time at which a given height occurs.

3. Catalan height as a calibrated geometric scale

Let $\mathcal{D}_n$ be the set of Dyck paths of semilength n and let $D_{n,h}$ denote the number whose height is at most h.

Proposition 3.1 (Bounded-height spectral formula [LIT])**.** *For $n \geq 1$ and $h \geq 1$,*

$$D_{n,h} = \frac{2}{h+2} \sum_{j=1}^{h+1} \sin^2\!\left(\frac{\pi j}{h+2}\right) \left(2\cos\frac{\pi j}{h+2}\right)^{2n}. \tag{3.1}$$

This is the transfer-matrix formula for the path graph on heights $0, 1, \ldots, h$. The corresponding Brownian-excursion scaling and moderate deviations are classical; see [8, 6, 9, 10].

Proposition 3.2 (Height phase transition [LIT])**.** *Fix $c > 0$ and put $h_n = \lfloor cn^\beta \rfloor$. Then the Catalan height threshold occurs at $\beta = 1/2$: for $\beta < 1/2$,*

$$\log \frac{D_{n,h_n}}{C_n} = -\frac{\pi^2}{c^2} n^{1-2\beta} + O(\log n), \tag{3.2}$$

whereas for $\beta > 1/2$ the ratio tends to one. At $\beta = 1/2$ the ratio has a non-degenerate Brownian-excursion limit distribution.

Warning 3.3 (Ambient rarity is not an arithmetic restriction)*.* The estimates in Proposition 3.2 describe the uniform Catalan ensemble. A deterministic arithmetic selector may choose a member of an exponentially small Catalan sector. No conclusion for M follows until an arithmetic property of the selector is proved.

A simple reflection estimate gives a finite bound useful for calibration.

Lemma 3.4 (Finite high-height tail [DED])**.** *For $1 \leq H \leq N$,*

$$\frac{\#\{D \in \mathcal{D}_N : \max D \geq H\}}{C_N} \leq (N+1) \exp\!\left(-\frac{H^2}{2N}\right). \tag{3.3}$$

Proof. A Dyck path reaching H is in particular an unrestricted bridge from 0 to 0 that reaches H. Reflection after its first visit to H injects such bridges into walks from 0 to $2H$, giving at most $\binom{2N}{N+H}$ possibilities. Since $C_N = (N+1)^{-1}\binom{2N}{N}$,

$$\frac{\#\{D : \max D \geq H\}}{C_N} \leq (N+1) \prod_{j=1}^{H} \frac{N-j+1}{N+j}. \tag{3.4}$$

Using $\log(1-u) \leq -u$ on the product yields (3.3). □

4. Occupation energy and the exact operator $J_H^* J_H$

For $q \geq 1$ define the ordinary-time Catalan occupation energy

$$\mathcal{E}_q(x, H) = \sum_{h=1}^{H} |M(x+h) - M(x)|^{2q}. \tag{4.1}$$

The compressed analogue is

$$\mathcal{E}_q^{\#}(m, L) = \sum_{u=1}^{L} |X_{m+u} - X_m|^{2q}. \tag{4.2}$$

These quantities should not be silently identified; ordinary time repeats heights at nonsquarefree integers.

Definition 4.1 (Dyadic Catalan energy)**.** For fixed $q \geq 1$, write $\mathrm{CE}_q^{\mathrm{dyad}}$ for the estimate

$$\mathcal{E}_q(x, x) \ll_{q,\varepsilon} x^{q+1+\varepsilon} \quad (x \geq 2). \tag{4.3}$$

Theorem 4.2 (Exact exponent ladder [DED])**.** *For every fixed $q \geq 1$,*

$$\mathrm{CE}_q^{\mathrm{dyad}} \quad \Longrightarrow \quad M(X) \ll_{q,\varepsilon} X^{(q+1)/(2q+1)+\varepsilon}. \tag{4.4}$$

Moreover,

$$\big[\mathrm{CE}_q^{\mathrm{dyad}} \text{ for every fixed } q\big] \quad \Longleftrightarrow \quad \mathrm{RH}. \tag{4.5}$$

Proof. Put $S_x(h) = M(x+h) - M(x)$ and $A = |S_x(x)|$. Since the path is one-Lipschitz, at least a constant multiple of A final values have size at least a constant multiple of A. Hence

$$A^{2q+1} \ll_q \mathcal{E}_q(x, x). \tag{4.6}$$

Estimate (4.4) follows for $M(2x) - M(x)$ and then by dyadic telescoping. If the hierarchy holds for all fixed q, the exponents decrease to $1/2$. Conversely, RH gives $|S_x(h)| \ll_\eta x^{1/2+\eta}$ uniformly for $x \leq x + h \leq 2x$, which implies (4.3) after choosing η in terms of q and ε. □

The quadratic member has an exact spectral structure. Put

$$a_n = \mu(x+n), \qquad 1 \leq n \leq H, \tag{4.7}$$

and let J_H be the cumulative-sum matrix,

$$(J_H a)_h = \sum_{n \leq h} a_n. \tag{4.8}$$

Then

$$\mathcal{E}_1(x, H) = \langle a, K_H a \rangle, \qquad K_H = J_H^* J_H, \tag{4.9}$$

with

$$(K_H)_{rs} = H + 1 - \max(r, s). \tag{4.10}$$

Proposition 4.3 (Exact occupation spectrum [DED])**.** *The eigenvalues of K_H are*

$$\lambda_k = \frac{1}{4\sin^2\!\left(\frac{(2k-1)\pi}{4H+2}\right)}, \qquad 1 \leq k \leq H. \tag{4.11}$$

In particular, for $k \ll H$,

$$\lambda_k \asymp \frac{H^2}{(2k-1)^2}. \tag{4.12}$$

Thus low Fourier-like modes are weighted much more strongly than generic high modes. This is the reason to retain prime colours or twisted sums until after the $J_H^* J_H$ weighting rather than projecting immediately to a scalar endpoint.

Let

$$A_{x,H}(\alpha) = \sum_{n=1}^{H} \mu(x+n) e^{2\pi i \alpha n}. \tag{4.13}$$

The sine eigenvectors of K_H give the deterministic comparison

$$\boxed{\mathcal{E}_1(x,H) < (H+1) \sup_{\alpha \in \mathbb{R}} |A_{x,H}(\alpha)|^2.} \tag{4.14}$$

5. What is already known analytically, and the first power frontier

Matomäki and Teräväinen proved cancellation of μ in every interval of length x^θ for every $\theta > 0.55$, with Ramaré extraction as a key new ingredient [16]. They also proved a uniform twisted estimate for $\theta > 3/5$; higher-uniformity results in all intervals at the threshold $5/8$ were later proved by Matomäki, Shao, Tao, and Teräväinen [17]. Combining the twisted estimates with (4.14) gives logarithmic anti-ballistic occupation-energy savings in their valid ranges.

Proposition 5.1 (Known anti-ballistic quadratic bounds [LIT]+[DED])**.** *Fix $\eta > 0$. The published twisted short-interval estimates imply bounds of the form*

$$\mathcal{E}_1(x,H) \ll_{\theta,\eta} \frac{H^3}{(\log x)^{2/3-2\eta}} \qquad (H \geq x^\theta,\ \theta > 3/5), \tag{5.1}$$

and, for every $A > 0$,

$$\mathcal{E}_1(x,H) \ll_{A,\eta} \frac{H^3}{(\log x)^A} \qquad (x^{5/8+\eta} \leq H \leq x^{1-\eta}). \tag{5.2}$$

These are genuine improvements over the trivial ballistic energy $O(H^3)$, but they are not power savings in H.

Lemma 5.2 (One-sided tent conversion [DED])**.** *For any integer-valued path $S(0) = 0$ with increments of modulus at most one, let $\Delta = \max_{0 \leq h \leq H} |S(h)|$. Then*

$$\frac{\Delta^3}{3} \leq \sum_{h=1}^{H} |S(h)|^2. \tag{5.3}$$

Proof. If $|S(h_0)| = \Delta$, one-Lipschitz continuity forces $|S(h_0 - t)| \geq \Delta - t$ for $0 \leq t \leq \Delta$. Summing the resulting squares gives (5.3). □

Conjecture 5.3 (First major open energy target [OPEN])**.** *There exist fixed $\theta, \delta > 0$ such that, uniformly in the initial point on dyadic ranges and for $H \asymp X^\theta$,*

$$\mathcal{E}_1(x,H) \ll H^{3-\delta} X^\varepsilon \tag{5.4}$$

for every $\varepsilon > 0$.

Proposition 5.4 (Propagation of a mesoscopic power saving [DED])**.** *If (5.4) holds, then*

$$M(X) \ll X^{1-\theta\delta/3+\varepsilon}. \tag{5.5}$$

Proof. Lemma 5.2 gives a window endpoint bound $O(H^{1-\delta/3} X^\varepsilon)$. Partition a dyadic interval into $O(X/H)$ windows and sum the endpoint increments. □

Remark 5.5 (Calibration). The Brownian target $\mathcal{E}_1 \ll H^{2+\varepsilon}$ is a long-term endpoint, not the first milestone. The first nonclassical analytic step is any fixed power saving in (5.4).

6. Why pointwise revelation is insufficient

Suppose a word of length m has a set F of already fixed signs. Define

$$s_F(t) = \sum_{\substack{j \le t \\ j \in F}} \sigma_j, \tag{6.1}$$

$$U_F(t) = \#\{j \le t : j \notin F\}. \tag{6.2}$$

Proposition 6.1 (Exact pointwise envelope [DED])**.** *For every prefix t,*

$$\max_{\textit{compatible completions}} |S_t| = U_F(t) + |s_F(t)|. \tag{6.3}$$

Consequently,

$$H_F^*(m) = \max_{t \le m} \big(U_F(t) + |s_F(t)|\big). \tag{6.4}$$

Proof. Every free sign in a fixed prefix can be chosen to reinforce the sign of $s_F(t)$; if the fixed sum is zero, choose all free signs with the same orientation. □

Corollary 6.2 (Near-total revelation barrier [NEG])**.** *A pointwise-revelation mechanism can force the envelope to $O(m^{1/2+\varepsilon})$ only if every relevant prefix contains at most $O(m^{1/2+\varepsilon})$ unrevealed positions, apart from cancellation already proved in the fixed part.*

Thus a proof based only on revealing $\omega(n)$ layers one position at a time approaches full knowledge of the Möbius word and risks becoming the desired Mertens estimate in disguise. Collective restrictions are required.

7. Consistent signed graphs and collective prefix geometry

A convenient universal model is a consistent signed graph on positions. If an edge uv carries $\eta_{uv} \in \{\pm 1\}$, impose

$$\varepsilon_v = \eta_{uv}\varepsilon_u. \tag{7.1}$$

On each connected component C there are deterministic labels $\lambda_v \in \{\pm 1\}$ and one free orientation s_C such that $\varepsilon_v = s_C\lambda_v$.

Definition 7.1 (Prefix charge)**.** For a component C define

$$Q_C(t) = \sum_{\substack{v \in C \\ v \le t}} \lambda_v. \tag{7.2}$$

Proposition 7.2 (Exact signed-graph envelope [DED])**.** *For every prefix t,*

$$\max_{\textit{compatible orientations}} |S_t| = \sum_C |Q_C(t)|, \tag{7.3}$$

and therefore

$$H_G^* = \max_t \sum_C |Q_C(t)|. \tag{7.4}$$

This identity converts an arithmetic relation graph into an exact geometric height problem. The next sections specialize it to multiplicative divisor components.

8. Adversarial multiplicative completions and signed displacement cuts

A separate project route begins with the finite cube of multiplicative squarefree sign completions. Fix the squarefree support and prime signs up to scale N. The true Möbius word is one vertex of this finite cube. For a completion x define

$$C_h(x) = \sum_{n \le N-h} x_n x_{n+h}, \qquad d_h(x) = C_h(x) - C_h(\mu). \tag{8.1}$$

Let $D(x) = \{n : x_n \neq \mu(n)\}$ on the squarefree support.

Theorem 8.1 (Exact signed-cut identity [DED])**.** *For each displacement h, there is a signed displacement graph G_h such that*

$$\frac{d_h(x)}{2} = \mathrm{Cut}_{G_h}(D(x)). \tag{8.2}$$

If $u_e \in \mathbb{F}_2^{\pi(N)-1}$ is the odd-prime parity form attached to an edge e, its sign satisfies

$$w_h(e) = (-1)^{|u_e|+h+1}. \tag{8.3}$$

The exact finite formulas convert every displacement test into a weighted parity-threshold condition on the prime-sign Boolean cube. Neighboring literature on random multiplicative functions, small partial sums, and sign patterns includes [2, 11, 15]; the specific finite extinction architecture is treated here only as a project construction, without a literature-wide priority assertion.

Experiment 8.2 (Finite extinction audit [COMP])**.** The project computations verified complete extinction of every non-Möbius multiplicative completion at several small finite scales, including $N = 25, 43, 53, 61, 71, 73$. This is finite evidence only. The general statement

$$\mathcal{B}_{H_0(N)}(N) = \{0\} \tag{8.4}$$

is not proved.

Warning 8.3 (Extinction route versus mean square). A Fejér or moving-window mean-square estimate can be formulated directly for the true μ sequence without the completion cube. Such an estimate may be mathematically useful, but it does not explain finite extinction and is not an internal continuation of that mechanism.

9. Collective divisor-cube capacity

We now introduce a collective arithmetic restriction that acts on whole multiplicative components rather than fixing signs position by position. Let

$$P = \prod_{p \in \mathcal{P}} p \tag{9.1}$$

be squarefree. Every squarefree integer has a unique representation

$$n = ad, \qquad d \mid P, \qquad (a, P) = 1. \tag{9.2}$$

The relations generated by the primes dividing P force every compatible sign assignment to have the form

$$\varepsilon(ad) = s_a \mu(d), \qquad s_a \in \{\pm 1\}. \tag{9.3}$$

Thus one bit remains for each P-free squarefree base a.

Fix a window $W = (A, T]$ and define

$$D_a(W) = \{d \mid P : A < ad \le T\}. \tag{9.4}$$

Let

$$E_a(W) = \#\{d \in D_a(W) : \mu(d) = +1\}, \tag{9.5}$$

$$O_a(W) = \#\{d \in D_a(W) : \mu(d) = -1\}. \tag{9.6}$$

Theorem 9.1 (Exact component capacity [DED]). *Within the component $S_a = \{ad : d \in D_a(W)\}$ the number of up-steps satisfies*

$$|B \cap S_a| \le \max(E_a, O_a). \tag{9.7}$$

Moreover, if only the multiplicative relations from P are imposed, then the exact maximum signed contribution over the whole window is

$$H_P^{\mathrm{free}}(A, T) = L(A, T) - 2\mathcal{B}_P(A, T), \tag{9.8}$$

where

$$L(A, T) = Q(T) - Q(A), \tag{9.9}$$

$$\mathcal{B}_P(A, T) = \sum_a \min(E_a, O_a). \tag{9.10}$$

Proof. In the component a, all signs equal $s_a\mu(d)$. If $s_a = +1$, the up-steps are the E_a divisors; if $s_a = -1$, they are the O_a divisors. Hence the maximal signed contribution of that component is $|E_a - O_a| = (E_a + O_a) - 2\min(E_a, O_a)$. Sum over the disjoint components. □

Remark 9.2 (Sign-sensitive fail gate). The deficit in (9.10) genuinely uses Möbius parity. Under the replacement $\mu \mapsto |\mu|$, all active divisors in a component have the same sign and the deficit disappears. Thus this capacity passes the sign-sensitivity test required of a nontrivial arithmetic layer.

For a multiplicative window $T = \lambda A$, define

$$g_{P,\lambda}(y) = \sum_{\substack{d|P \\ y<d\le\lambda y}} \mu(d). \tag{9.11}$$

Theorem 9.3 (Exact asymptotic capacity ratio for fixed P [DED]). *For fixed squarefree P and $\lambda > 1$,*

$$H_P(A, \lambda A) = \sum_{\substack{a \text{ squarefree} \\ (a,P)=1}} \left| g_{P,\lambda}(A/a) \right|. \tag{9.12}$$

Furthermore,

$$H_P(A, \lambda A) = \delta_P A I_P(\lambda) + O_P(\sqrt{A}), \tag{9.13}$$

where

$$\delta_P = \frac{6}{\pi^2} \prod_{p|P} \frac{p}{p+1}, \tag{9.14}$$

$$I_P(\lambda) = \int_{1/\lambda}^{\infty} \left| \sum_{\substack{d|P \\ y<d\le\lambda y}} \mu(d) \right| \frac{dy}{y^2}. \tag{9.15}$$

Consequently the surviving ballistic ratio is

$$\boxed{\rho_P(\lambda) = \frac{1}{\lambda - 1} \left(\prod_{p|P} \frac{p}{p+1} \right) I_P(\lambda).} \tag{9.16}$$

Proof. For each base a, the signed component sum is $s_a g_{P,\lambda}(A/a)$, proving (9.12). The count of squarefree a coprime to fixed P has density δ_P and an $O_P(\sqrt{A})$ elementary squarefree-counting error. Since $g_{P,\lambda}$ is a finite step function, summation over its finitely many threshold intervals gives (9.13). Divide by $Q(\lambda A) - Q(A) = (6/\pi^2)(\lambda - 1)A + O(\sqrt{A})$. □

Example 9.4. For $P = 30$ and $\lambda = 2$, the opposite-parity divisors 5 and 6 already force the full deficit and one obtains

$$\rho_{30}(2) = \frac{8}{9}. \tag{9.17}$$

For $P = 210$ the corresponding exact finite-divisor calculation gives

$$\rho_{210}(2) = \frac{5}{6}. \tag{9.18}$$

These are constant-factor contractions, not exponent contractions.

10. Prime insertion, monotonicity, and the entropy–capacity barrier

Set

$$F_{P,\lambda}(x) = \sum_{d|P} \mu(d) \mathbf{1}_{(1/d,\lambda/d]}(x) \tag{10.1}$$

and

$$c(P) = \prod_{p|P} \frac{p}{p+1}. \tag{10.2}$$

Then

$$\rho_P(\lambda) = \frac{c(P)}{\lambda - 1} \|F_{P,\lambda}\|_{L^1(0,\infty)}. \tag{10.3}$$

Proposition 10.1 (Delayed anti-copy under prime insertion [DED])**.** *If $p \nmid P$, then*

$$F_{Pp,\lambda}(x) = F_{P,\lambda}(x) - F_{P,\lambda}(px). \tag{10.4}$$

Consequently,

$$\frac{p-1}{p+1} \rho_P(\lambda) \le \rho_{Pp}(\lambda) \le \rho_P(\lambda). \tag{10.5}$$

Proof. Split the divisors of Pp into d and pd with $d \mid P$, using $\mu(pd) = -\mu(d)$. For the upper bound, apply the triangle inequality to (10.4); for the lower bound apply the reverse triangle inequality and use $\|F_{P,\lambda}(p\,\cdot)\|_1 = p^{-1}\|F_{P,\lambda}\|_1$. Multiplication by $c(Pp) = p(p+1)^{-1}c(P)$ gives (10.5). □

Corollary 10.2 (Structural lower barrier [DED])**.** *For every finite squarefree P,*

$$\prod_{p|P} \frac{p-1}{p+1} \le \rho_P(\lambda) \le 1. \tag{10.6}$$

In particular,

$$\rho_P(\lambda) \ge \frac{c(P)^2}{\zeta(2)}. \tag{10.7}$$

Proof. Iterate (10.5) from $P = 1$. Since

$$\prod_{p|P} \frac{p-1}{p+1} = c(P)^2 \prod_{p|P} \left(1 - \frac{1}{p^2}\right), \tag{10.8}$$

and the finite Euler product is at least $1/\zeta(2)$, (10.7) follows. □

The exact gain from a newly inserted prime can also be isolated. Define

$$J_{P,p}(\lambda) = \int_{\{F_{P,\lambda}(x)F_{P,\lambda}(px)>0\}} \min\{|F_{P,\lambda}(x)|, |F_{P,\lambda}(px)|\}\, dx. \tag{10.9}$$

Then the identity $|u-v| = |u| + |v| - 2\min(|u|,|v|)$ on the same-sign region gives

$$\rho_P(\lambda) - \rho_{Pp}(\lambda) = \frac{2p}{p+1}\frac{c(P)}{\lambda-1} J_{P,p}(\lambda). \tag{10.10}$$

Thus the layer synergy is exactly the same-sign overlap between a divisor kernel and its multiplicative dilation.

For the primorial $P_z = \prod_{p\le z} p$, Mertens' product theorem gives

$$c(P_z) \sim \frac{\zeta(2)e^{-\gamma}}{\log z}. \tag{10.11}$$

Hence (10.7) implies the structural lower scale

$$\rho_{P_z}(\lambda) \gtrsim \frac{\zeta(2)e^{-2\gamma}}{(\log z)^2}. \tag{10.12}$$

Warning 10.3 (Entropy–capacity tradeoff [NEG]). Driving ρ_{P_z} toward zero by taking $z\to\infty$ also drives the density of P_z-free base components toward zero. The number of independent orientations s_a in a macroscopic interval is of order $A/\log z$ in the usual sieve range, so the resulting number of componentwise free completions is only

$$2^{O(A/\log z)} = 2^{o(A)} \qquad (z\to\infty). \tag{10.13}$$

Thus this within-component mechanism cannot simultaneously produce sublinear confinement and retain 2^{cA} independent component orientations with fixed $c>0$.

11. Static Ramaré does not couple divisor components

The Catalan programme uses the exact Ramaré colouring

$$\mu(n) = -\sum_{\substack{p\in\mathcal{P}\\ p\mid n}} \frac{\mu(n/p)}{\nu_{\mathcal{P}}(n)}, \qquad \nu_{\mathcal{P}}(n) = \#\{p\in\mathcal{P} : p\mid n\}, \tag{11.1}$$

for squarefree n having at least one prime factor in the band $\mathcal{P}$. This weighted extraction is a central analytic device in the short-interval work of Matomäki and Teräväinen [16].

Proposition 11.1 (Static component decomposition [NEG])**.** *Fix a finite prime set $\mathcal{P}$ and let $P=\prod_{p\in\mathcal{P}} p$. Form the graph on squarefree integers with an edge between n and n/p whenever $p\in\mathcal{P}$ divides n. Its connected components are exactly*

$$\mathcal{C}_a = \{ad : d\mid P\}, \qquad (a,P)=1, \tag{11.2}$$

with the obvious truncation if a finite cutoff is imposed. Every sign assignment compatible with all static prime-deletion relations has one free bit per component and has the form (9.3)*.*

Proof. Deleting primes from $\mathcal{P}$ changes only the divisor d and leaves the P-free part a unchanged. Conversely, any two divisors of P can be connected by successive insertion or deletion of primes from $\mathcal{P}$. Möbius parity changes once per prime insertion, yielding (9.3). □

Corollary 11.2 (No static inter-component gain [NEG])**.** *A reweighting of the static identities* (11.1) *alone cannot correlate the independent orientations s_a of distinct components. Any gain beyond Theorem 9.3 must use additional information, such as the causal order/cutoff, an analytic recombination estimate, or another relation coupling different bases.*

This no-go is compatible with the "redundant prime lift" already present in the Catalan notebook: subtracting the exact prime-shell identity at two excursion endpoints reproduces the same Mertens prefix for every prime and gives no estimate by averaging over primes.

12. The causal Ramaré prefix lift and the open inter-component frontier

For an interval beginning at a, retain the prime colour until after the prefix has been formed. Define the coloured currents

$$C_p(u) = -\sum_{\substack{a<n\le a+u\\ p\mid n,\ \mu(n)^2=1\\ \nu_{\mathcal{P}}(n)>0}} \frac{\mu(n/p)}{\nu_{\mathcal{P}}(n)} \tag{12.1}$$

and the rough remainder

$$R_{\mathcal{P}}(u) = \sum_{\substack{a<n\le a+u\\ \nu_{\mathcal{P}}(n)=0}} \mu(n). \tag{12.2}$$

Then every prefix satisfies the exact vector identity

$$\boxed{M(a+u) - M(a) = R_{\mathcal{P}}(u) + \sum_{p\in\mathcal{P}} C_p(u).} \tag{12.3}$$

Unlike the static component relations, the functions $u \mapsto C_p(u)$ retain the causal truncation and can be passed through the occupation operator $J_H^* J_H$ before scalar projection.

Conjecture 12.1 (Weak causal prime-incidence power saving [OPEN])**.** *There exist admissible growing prime bands and fixed $\delta > 0$ such that the signed recombination in* (12.3)*, after low-mode $J_H^* J_H$ weighting and with the rough term retained, implies*

$$\mathcal{E}_1(x, H) \ll H^{3-\delta} X^{\varepsilon} \tag{12.4}$$

uniformly on a nontrivial polynomial window range.

Remark 12.2 (Long-term Brownian frame)*.* The stronger square-function target of order $H^{2+\varepsilon}$ is retained as the Brownian endpoint of the programme, not as the first milestone. The finite-power estimate (12.4) is the calibrated first open target.

13. Catalan rank and collective capacity constraints

A generic collective restriction on up-step positions has the form

$$|B \cap S| \le c(S). \tag{13.1}$$

In the Catalan matroid description, a Dyck basis with ordered up-step positions $b_1 < \cdots < b_n$ satisfies

$$b_i \le 2i - 1 \qquad (1 \le i \le n). \tag{13.2}$$

The project notebook develops a universal rank-to-height conversion for restrictions of this type.

Proposition 13.1 (Exact layer drop in a Catalan minor [DED])**.** *Suppose a Catalan completion class has current free prefix set E_t and matroid rank $r_M(E_t)$. A new layer fixes previously free sets P_+ upward and P_- downward, producing a new minor M'. Define*

$$\Delta_t = r_M(E_t) - |P_+ \cap I_t| - r_{M'}(E_t \setminus (P_+ \cup P_-)). \tag{13.3}$$

Then $\Delta_t \ge 0$ and the maximal compatible prefix height drops by

$$h^*_{\mathrm{old}}(t) - h^*_{\mathrm{new}}(t) = 2\Delta_t. \tag{13.4}$$

The divisor-cube capacity in Section 9 supplies an explicit arithmetic source of a collective deficit; the unresolved task is to obtain a growing family whose joint deficit acts at mesoscopic scales without spending essentially all remaining entropy.

14. Sign constraints and exact ballot suppression

The project chat on Mertens sign changes also gives a useful negative calibration. Fix the actual zero positions of μ and require a candidate ternary word to have partial sums with the same sign pattern as M. In the initial example through $n = 20$, after the forced touch at zero and the first negative steps, the remaining problem is a one-sided ballot problem.

Proposition 14.1 (Closed count for the initial sign-constrained model [DED])**.** *Let n denote the number of nonzero steps in the model described above. For $n \geq 4$, the number of admissible words is*

$$R(n) = \binom{n-3}{\lfloor (n-3)/2 \rfloor}. \tag{14.1}$$

Thus, among the 2^n unrestricted sign words,

$$\frac{R(n)}{2^n} \sim \frac{1}{8}\sqrt{\frac{2}{\pi(n-3)}}. \tag{14.2}$$

Proof. After the finite forced prefix, reflect the negative walk to a walk starting at height two and require it to avoid zero. The reflection principle gives a difference of binomial coefficients for each endpoint. Summing over admissible endpoints telescopes to the central binomial expression in (14.1). Stirling's formula gives (14.2). □

Remark 14.2 (Interpretation [NEG]). In this model, sign/crossing information suppresses the ambient word count only by a polynomial factor. It is standard ballot geometry, not yet a Möbius-specific cancellation mechanism.

15. Ordinal-prime transform: exact information and coercive negative excursions

A complementary project assigns ordinal primes separately to positive and negative nonzero Möbius occurrences, producing a sequence $S(n)$ and partial sum $SS(n)$. Its endpoint theory is exact but largely determined by the pair $(M(n), Q(n))$. After subtracting the deterministic zero contribution, the remaining term is a signed block of $|M(n)|$ consecutive primes. Thus its endpoint RH criteria are reformulations rather than independent constraints.

The transform nevertheless contains a useful pathwise asymmetry on negative excursions. Let

$$R(q) = M(q_q) \tag{15.1}$$

be the compressed walk in the notation of the ordinal project. A complete negative excursion has depth

$$H = -\min R(q). \tag{15.2}$$

For each level $-k$, let u_k, v_k be its first and last visits and write $v_k - u_k = 2r_k$.

Theorem 15.1 (Coercive negative-excursion energy [DED])**.** *The ordinal-prime level-return functional $\mathcal{A}$ defined by the positive prime-separation sums satisfies*

$$\mathcal{A} \geq \frac{H^3 - H}{3}. \tag{15.3}$$

More sharply, if

$$G(k) = \inf_{m \geq 2}(p_{m+k} - p_m), \tag{15.4}$$

then

$$\mathcal{A} \geq \sum_{k=1}^{H-1} (H-k)G(k). \tag{15.5}$$

Warning 15.2 (What the cubic lower bound does not do). Theorem 15.1 becomes an upper bound on the excursion depth only after an independent upper estimate for $\mathcal{A}$ is proved. No such estimate at the required scale is presently supplied by the transform itself.

16. The analytic α programme as a calibrated companion

The ordinal programme also produces Dirichlet-type functions whose convergence boundaries encode the zeta zero frontier. Let

$$\Theta = \sup_{\zeta(\rho)=0} \Re\rho. \tag{16.1}$$

The source notebook proves, for its explicitly defined α and paired term δ, the abscissa identity

$$\sigma_c(\alpha) = \sigma_c(\delta) = 1 + \Theta. \tag{16.2}$$

It also isolates a component $B(s)$ whose poles occur at $1+\rho$ with multiplicity inherited from the zeros of ζ.

Warning 16.1 (Pole transfer boundary). The poles of the explicit B component do not by themselves prove corresponding poles for the full α or δ. A relatively smaller Mellin remainder can still generate a logarithmic branch singularity. Continuation of the full object requires an independent uniform estimate.

An exact energy version is particularly clean. If

$$\mathcal{I}(\sigma) = \int_1^\infty \left(SS(x) - \lfloor x \rfloor + Q(x)\right)^2 x^{-2\sigma-1}\, dx, \tag{16.3}$$

then the project establishes an exact comparison with a weighted Mertens mean square and consequently

$$\text{RH} \quad \Longleftrightarrow \quad \mathcal{I}(\sigma) < \infty \text{ for every } \sigma > 3/2. \tag{16.4}$$

This is a useful calibration of exponent strength, not an independent proof mechanism.

17. Divisibility surfaces and p-adic shell coordinates

The divisibility-surface programme introduces

$$F(n,r) = \sum_{\substack{a \mid n \\ a > r}} \mu(n/a) \tag{17.1}$$

and a finite square matrix B_N built from these values. One of its exact structural identities is the explicit inverse

$$(B_N^{-1})_{r,n} = \mathbf{1}_{n|r+1} - \mathbf{1}_{n|r}. \tag{17.2}$$

This supplies a common coordinate system for Möbius, Liouville, squarefree projections, and additive shifts. The construction is structural: by itself it does not create the missing cancellation needed for ordinary fixed-shift Chowla or (1.4).

A related p-adic shell factorization writes suitable Liouville correlations as a valuation-parity factor times a p-free part. The special triadic centering can eliminate a constant local channel, but the remaining additively modulated p-free channels still require cancellation. This yields another general methodological rule: local martingale orthogonality is not arithmetic independence.

Warning 17.1 (Single-prime shell no-go). The single-prime shell route does not provide a universal base-independent contraction. Exact shell decompositions are retained as coordinates, but closure of their tails requires cancellation of essentially the original Chowla/Mertens type. They are not used here as a proof shortcut.

18. One-way Catalan calibration of hypothetical off-critical zeros

This section records a consequence, not a proof strategy. Suppose ζ has a zero $\rho = \beta + i\gamma$ with $\beta > 1/2$. The Mellin representation of $1/\zeta$ implies that for every $\varepsilon > 0$, $M(x)$ cannot be $O(x^{\beta-\varepsilon})$; see [25]. Thus there are arbitrarily large points with Mertens displacement at that exponent scale.

Take such a compressed point m, let $z < m$ be the last return to zero, reflect the open meander if necessary, and close it by adding $|X_m|$ downward steps. This gives a canonical Dyck path of some semilength $N_m \le m$ and height at least $|X_m|$. Combining this with Lemma 3.4 yields the following calibration.

Proposition 18.1 (Off-critical zero forces rare Catalan selections [DED]). *If $\zeta(\beta + i\gamma) = 0$ with $\beta > 1/2$, then for every $\varepsilon > 0$ there are arbitrarily large compressed times for which the canonical closed Mertens meander belongs to a Catalan height sector of relative size at most*

$$\exp\Big(-m^{2\beta-1-2\varepsilon+o(1)}\Big). \tag{18.1}$$

Warning 18.2 (Not a selector theorem). Proposition 18.1 is one-way. It says what an off-critical zero would force the arithmetic selector to do. Excluding all such rare sectors is essentially the unresolved Mertens/RH problem unless exclusion is obtained from an independent arithmetic mechanism. No low-height anomaly is inferred from the reflected zero $1 - \beta$.

The functional equation pairs β with $1 - \beta$, while $1/2$ is its unique fixed point. Independently, $1/2$ is the Catalan height-transition exponent of Proposition 3.2. This coincidence is a useful structural calibration, not an identity between zeta zeros and Catalan species.

19. Project computations on the full Möbius data set

The following experiments belong to the project data analysis. They are finite statements and are not used as asymptotic premises.

Experiment 19.1 (Excursion census through 455,052,511 [COMP]). A blockwise scan of the binary Möbius data through

$$N = 455{,}052{,}511 \tag{19.1}$$

reported the following values:

quantity	measured value
nonzero squarefree steps	276,638,778
$M(N)$	−5414
returns of the compressed walk to zero	53,078
real sign changes between excursions	26,720
last ordinary-time return	420,059,551
last compressed-time return	255,365,638
open compressed tail	21,273,140
mean completed-excursion gap	4811.139041
maximum completed-excursion gap	33,409,232

The same audit computed a cumulative Catalan-compatible path count with $\log_{10}$ fraction approximately −77,335.1889795 relative to all unconstrained nonzero sign words. This enormous ambient entropy loss does not by itself imply height confinement, by Warning 3.3.

Experiment 19.2 (Finite square-root envelope audit [COMP])**.** An exhaustive scan over every integer

$$1000 \le x \le 455{,}052{,}511 \tag{19.2}$$

found

$$|M(x)| \le 0.472203\sqrt{x} \tag{19.3}$$

throughout that finite range, with the maximum normalized value attained at $x = 2803$. On the same squarefree support, 25 random-sign controls had mean normalized maximum 1.8972 and standard deviation 0.3384; none was below the Möbius value.

Warning 19.3 (Finite envelope is not an $O(\sqrt{x})$ theorem)*.* Experiment 19.2 is a finite computation only. It neither proves nor suggests that the false Mertens conjecture should hold asymptotically. The correct RH-scale benchmark remains (1.4).

Experiment 19.4 (Catalan height-percentile audit [COMP])**.** For every completed excursion, the project code compares the observed maximum with all primitive Catalan excursions having the same compressed length and fixed excursion sign. Because the height distribution is discrete, the calibrated null statistic is the mid-rank

$$u_r = \mathbb{P}(H < H_r) + \frac{1}{2}\mathbb{P}(H = H_r), \tag{19.4}$$

whose expectation under the uniform Catalan null is exactly $1/2$. An earlier raw geometric average of cumulative probabilities was about 79.3%, but that raw statistic is not uniform under the discrete null and is retained only as a diagnostic. The mid-rank, not the raw geometric percentile, is the correct calibrated observable.

20. A unified master spectrum for partial arithmetic information

Let $\mathcal{P}_r(m)$ be a class of nearest-neighbour paths compatible with the arithmetic information revealed through layer r, together with any explicitly imposed return, sign, endpoint, or meander constraints. Define

$$\mathfrak{Z}_{r,m}(v,z,u) = \sum_{Y \in \mathcal{P}_r(m)} v^{\mathcal{R}_m(Y)} z^{Y_m} u^{\max_{0\le j\le m}|Y_j|}, \tag{20.1}$$

where $\mathcal{R}_m(Y)$ counts returns to zero.

This single polynomial retains three quantities that total completion counts lose: return structure, endpoint displacement, and maximum height. The corresponding high-sector count is

$$N_r^{\ge H}(m) = \#\{Y \in \mathcal{P}_r(m) : \max_{j\le m}|Y_j| \ge H\}. \tag{20.2}$$

Remark 20.1 (What counts as genuine progress)*.* A reduction in $|\mathcal{P}_r(m)|$ is not sufficient. A useful partial theorem must preferentially reduce $N_r^{\ge H}(m)$ or an occupation-energy analogue while leaving a nontrivial family of completions alive. This criterion prevents both full-revelation tautologies and entropy-only claims.

21. Priority research programme

The preceding exact results narrow the next steps considerably.

Question 21.1 (Causal inter-component capacity)**.** Can the causal prime-incidence lift (12.3) produce a proved family of constraints

$$|B \cap S| \le c(S) \tag{21.1}$$

that couples distinct divisor bases a and yields a power-size cumulative prefix deficit before near-total revelation?

Question 21.2 (Low-mode Ramaré frame)**.** Can one bound the low eigenmodes of $J_H^* J_H$ after Ramaré decomposition without summing absolute values of all cross-colour correlations, and obtain (12.4)?

Question 21.3 (Extinction-to-height transfer)**.** Can first-witness signed-cut extinction cells be shown to preferentially remove high Catalan sectors, rather than merely reducing the total number of multiplicative completions?

Question 21.4 (Ordinal cubic energy upper bound)**.** Is there an independent upper bound on the negative-excursion functional $\mathcal{A}$ strong enough, together with (15.3), to yield a nontrivial depth estimate?

Question 21.5 (Filtered shadow restoration)**.** Björner's squarefree simplicial complex has Mertens as an Euler-characteristic quantity [4]. Can a filtered shadow or persistent-homology inequality force opposite-parity mass to enter within a bounded future multiplicative window, rather than merely exist somewhere in the static complex?

The priority order is therefore

$$\boxed{\begin{aligned}&\text{causal arithmetic relation} \Longrightarrow \text{ collective prefix/low-mode deficit}\\ &\qquad\qquad\qquad\qquad\Longrightarrow \mathcal{E}_1 \ll H^{3-\delta}\\ &\qquad\qquad\qquad\qquad\Longrightarrow \text{ global exponent improvement.}\end{aligned}} \tag{21.2}$$

The Brownian $H^{2+\varepsilon}$ scale and the all-q RH-complete hierarchy remain later stages.

22. Negative results and circularity audit

The following barriers are retained because they prevent the programme from cycling through equivalent formulations.

Proposition 22.1 (Consolidated no-go list [NEG])**.** *The present collection establishes or records the following limitations.*

(i) Ambient Catalan rarity does not restrict a deterministic arithmetic selector.
(ii) Pointwise revelation cannot force the RH scale while a macroscopic number of independent signs remain free.
(iii) Any fixed finite set of multiplicative factor relations leaves a positive linear local capacity in sufficiently short relative windows.
(iv) Static finite-band Ramaré relations have exactly the same $aD(P)$ connected components as the divisor-cube capacity and do not couple their free orientations.
(v) A moving prime cutoff can reduce the capacity ratio, but the componentwise free entropy simultaneously loses positive density.
(vi) Scalar short-interval chord flatness alone does not rule out gentle high mountains; occupation energy contains strictly more path information.
(vii) The ordinal endpoint transform is a deterministic function of (M, Q) and supplies no independent endpoint constraint.
(viii) The explicit poles of a component of the α decomposition do not automatically transfer to the full function without a continuation estimate.
(ix) The divisibility surface and p-adic shell coordinates are exact changes of structure but do not themselves create a spectral or cancellation gap.
(x) The single-prime shell route is blocked unless one proves cancellation in the remaining p-free modulated channels.

23. Conclusions

The most useful common object across the project is not a single transform but the squarefree Mertens path equipped with three compatible structures: its Catalan return geometry, its multiplicative completion space, and its occupation-energy operator. The Catalan state space is exact; the arithmetic selector is the missing mechanism. The divisor-cube theorem supplies a first explicit Möbius-specific collective capacity and quantifies exactly how finite prime layers lower the ballistic constant. Its lower barrier and entropy tradeoff show why the same mechanism cannot simply be iterated to the RH scale. The static Ramaré no-go then identifies where genuinely new information must enter: through causal truncation and analytic recombination of prime-coloured currents, or through another inter-component relation.

The calibrated first analytic objective is therefore not RH and not the Brownian endpoint. It is the fixed-power occupation saving (5.4). The calibrated first combinatorial objective is a collective capacity or extinction statement that preferentially removes high sectors while retaining substantial freedom. These two objectives are close enough to interact through the low modes of $J_H^* J_H$, yet weak enough not to be merely restatements of (1.4).

Notebook history

Version 1, 15 September 2026. Consolidated the Catalan–Mertens excursion programme, the occupation-energy/J^*J route, adversarial completions and signed-cut extinction, the ordinal-prime and α programmes, divisibility-surface/p-adic coordinates, the project excursion and sign-change experiments, and the new divisor-cube collective-capacity analysis. Added the monotone capacity ratio ρ_P, the entropy–capacity barrier, and the static Ramaré component no-go. No literature-wide novelty claim is made for these notebook deductions.

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